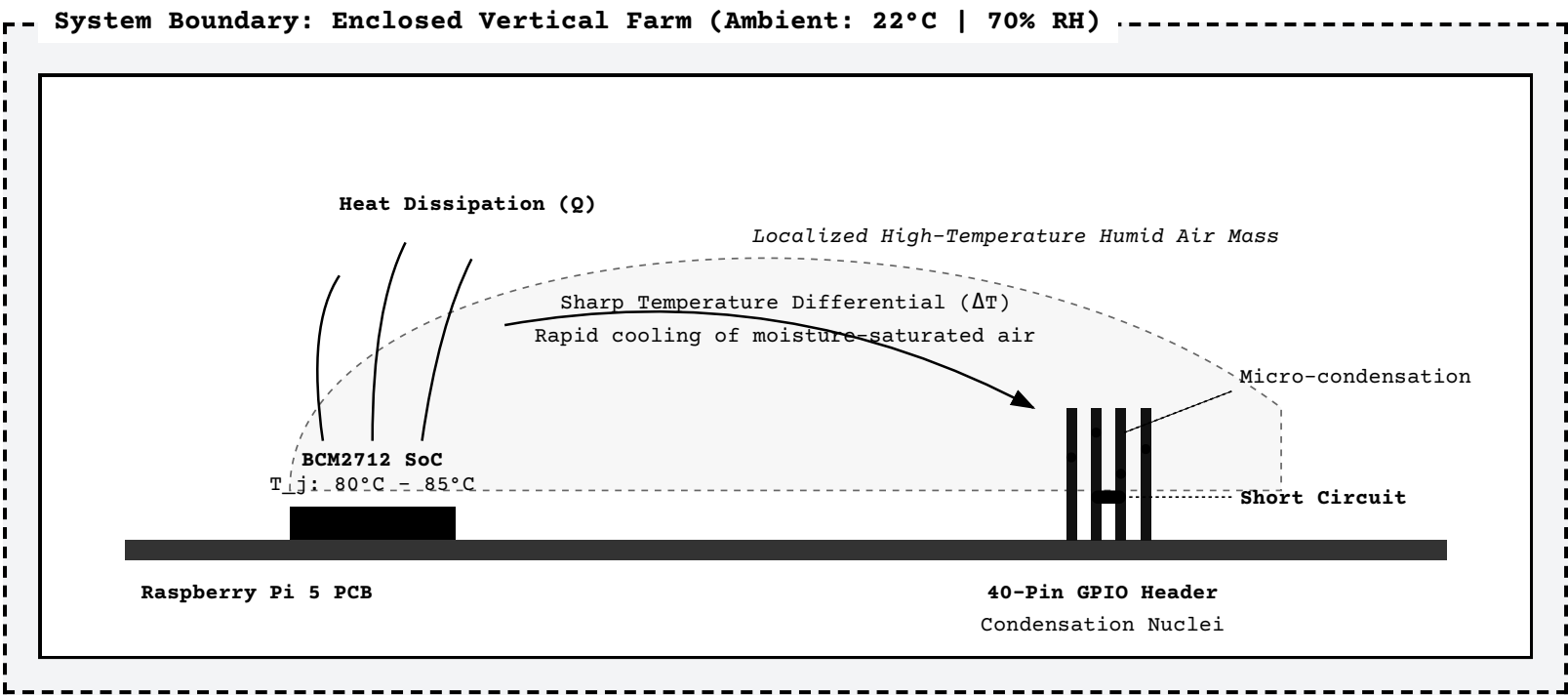


FIGURE 1: CRITICAL FAILURE ANALYSIS

Thermodynamic Cascade & Micro-Condensation on BCM2712 Architecture



PHASE ANALYSIS: THERMODYNAMIC CASCADE

PHASE I	PHASE II	PHASE III
Thermal Generation Primary Component: Raspberry Pi 5 (BCM2712 SoC) selected for computational benchmarks. Thermal Density: Operates at a significantly higher TDP than predecessors. T_J = 80°C to 85°C	Dew Point Acceleration Environment: High-humidity enclosed vertical farm (Ambient: 22°C, 70% RH). Mechanism: Localized, intense heat creates a sharp temperature differential (ΔT). Effect: Accelerated localized dew point due to thermal profiling interactions with humid air.	Catastrophic Failure Nucleation Site: Exposed silicon/metal of the 40-pin GPIO header. Manifestation: Immediate formation of micro-condensation. RESULT: HARDWARE DESTRUCTION VIA SHORT CIRCUIT